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cd2es: Converting climate data to energy system input data

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Software

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Summary

The tool *cd2es* converts downscaled climate projections from the cordex initiative (WCRP, 2024) to energy system input data. The tool is written in python and uses snakemake for workflow management. *cd2es* can calculate climate projection-based capacity factors for wind power, solar photovoltaics and concentrated solar power, as well as impacts of climate change on the availability of thermal power plants, hydro power and electricity demand. In contrast to existing tools, *cd2es* can use a variety of different climate models provided by WCRP (2024) instead of focusing on historic weather data. To avoid biases in the climate projections, a bias-adaption with era5-reanalysis data can be performed. The download of both cordex and era5 data can be done automatically within the workflow of *cd2es*. The different time series are either aggregated to country level or to a user specific geometry which the user provides in a geojson file. The outputs are time series with an hourly resolution which are saved in csv files for easy implementation into different energy system models.

Statement of need

To project future climate developments, climate scientists have developed Global Circulation Models (GCMs), which model the physical properties of the atmosphere and oceans. These are often downscaled for improving spatial resolution (Jacob et al., 2014). Even if the warming of the earth is limited to 2°C, GCMs project a significant change in climate variables such as temperature and precipitation (Dosio & Fischer, 2018). Yet, as many components of the energy system depend on climate variables such as temperatures, climate change should be considered when planning future energy systems. It is therefore important to convert climate projections from GCMs into input date for energy system optimization models (ESOMs), which are commonly used in energy systems planning to handle the complexity of large-scale energy systems such as the European electricity system Plaga & Bertsch (2023).

However, to use climate projections in energy system optimization models, energy system input data like electricity demand factors have to be derived from the climate variables like temperature. There are only a limited number of tools available which convert climate variables to energy system input data. An easy way to calculate solar and wind capacity factors from historical reanalysis data is renewables.ninja (Pfenninger & Staffell, 2016; Staffell & Pfenninger, 2016), but here, neither the underlying data nor the methods are exposed, thus it is not possible to use climate projections as input data. The python library pvlib (Holmgren et al., 2018) allows for a detailed calculation of solar capacity factors, yet is limited to historic data input and the calculation of solar capacity factors. Pypsa/atlite (Hofmann et al., 2021) converts weather data to energy system data, allowing for the calculation of wind, solar and hydro capacity factors and demand time series. Yet, it cannot account for climate projections and does not calculate climate influences on thermal power plants. Formayer et al. (2023) provide a European wide data set for temperature, radiation, wind power and hydro power based on

future climate projections. However, they only provide results for one climate model and no ready to use code to enlarge the findings to other climate models. Furthermore, they are not calculating impacts on solar and thermal power plants or demand. In summary, there is a lack of open source tools to include climate projections into energy system planning in a comprehensive matter.

The *cd2es* tool aims to provide a software for the automated download of climate projections and the following conversion to energy system input data. It supports the calculation of wind capacity factors, solar photovoltaics, concentrated solar power, availability of thermal power plants, hydro power and electricity demand based on cordex climate data (WCRP, 2024). The tool can process a variety of different climate models hosted on cordex. It also allows for an optional bias correction of the climate data based on era5-reanalysis data (Muñoz Sabater, 2019). The output of *cd2es* are csv files which provide capacity factors for the different technologies in hourly resolution for a user-chosen geography. Therefore, they can be easily included in different energy system models. The *cd2es* tool was used in (Plaga & Bertsch, 2022) and is currently in use in two research projects funded by the German Federal Ministry for Economic Affairs and Climate Action, StEAM and REWARDS (Härtel, 2024).

Software dependencies

The software is written in python and uses the workflow management tool snakemake (**?**). For processing climate data from cordex, the open-source tool *cdo* (Schulzweida, 2020) is included in the software. On Windows, additionally the open-source tool *wsl* is used to install a linux subsystem, which is necessary for *cdo* to run on Windows (microsoft, 2024).

Methods for converting climate variables to energy system input data

Most conversion methods are based on the review of Plaga & Bertsch (2023) and described in detail in the documentation of the tool *cd2es*. However, we will also include a short overview here.

Bias adaption

The climate data is bias adapted using a quantile delta mapping approach (Cannon et al., 2015).

Wind power

The wind speed $v(h_0)$ is first interpolated from height in the data h to hub height using h

$$v(h) = v(h_0)\left(\frac{h}{h_0}\right)^{\frac{1}{7}}.$$

Then capacity factors can be derived via a standardized production curve

$$cf_{\text{wind, single}}(v) = \begin{cases} 0, & v < v_{\text{in}}, \\ \frac{v^3 - v_{\text{in}}^3}{v_{\text{r}}^3 - v_{\text{in}}^3}, & v_{\text{in}} \leq v < v_{\text{r}}, \\ 1, & v_{\text{r}} \leq v < v_{\text{out}}, \\ 0, & v \geq v_{\text{out}}, \end{cases}$$

with the cut-in velocity v_{in}, the rated velocity v_{r} and the cut-out velocity v_{out} (van der Wiel et al., 2019). According to (Staffell & Pfenninger, 2016), we smooth the production curve with a gaussian filter to account for multiple turbines.

Solar photovoltaics

Photovoltaic cells are influenced by climate variables in two ways: the solar irradiance G influences the available incoming energy, while the outdoor temperature T influences the cell's efficiency. The *cd2es* tools supports three different models for calculating photovoltaic time series, as given in Jerez et al. (2015).

Availabiltiy of thermal power plants

As thermal power plants need cooling water, their availability decreases if temperatures exceed a certain threshold. *cd2es* distinguishes between once-through cooled plants and closed-loop cooled plants. For once-through plants, not only the temperature but also available water is relevant for availability calculations. The availability of the plants follows piecewise linear equations based on the temperature, which were implemented as described in (Abdin et al., 2019).

Hydropwer

To calculate hydro power output, historic runoff at the hydro power plants' locations are evaluated. It is assumed, that power plants reach their installed capacity P_0 at the average historic runoff $\bar{Q}_{\text{hist}}$ at their location (this assumption can be modified by multiplying a factor a). Then, future hydro power of one power plant $P(t)$ can be calculated with

$$P(t) = Q(t) \cdot \frac{P_0}{a \cdot \bar{Q}_{\text{hist}}}$$

using the linear relation between runoff and hydro power production and the future runoff $Q(t)$ (Schlott et al., 2018).

Demand

To calculate future demand, a quadratic regression is performed between historic temperatures and historic demand data as in (Zhang & Ayyub, 2020). The parameters derived here are then used to scale future demand time series.

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